Sahaj Bhadja

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Summary

Software developer with 2+ years building **Python data pipelines, research software, analytical APIs, and metadata-driven applications**, specializing in complex datasets, troubleshooting, testing, documentation, and collaborative development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built a multi-tenant research platform with **normalized metadata schemas, row-level isolation, indexed queries, and validation**, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ daily users.
- Developed FastAPI data workflows with **explicit job states, asynchronous processing, failure isolation, monitoring, and production troubleshooting**, tripling throughput for 1K+ analytical jobs while consistently maintaining 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal data, applying **structured preprocessing, validation, concurrent execution, and error handling** while reducing analytical turnaround time by 60%.
- Developed React and Flask interfaces supporting 150+ instructors and 10K+ daily requests, translating **user workflows into APIs, analytical controls, history views, testing procedures, and documented release changes**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python and PyTorch pipelines processing 100K+ driving trajectories, improving planning 18% through **time-series preprocessing, feature construction, simulation analysis, reproducible evaluation, benchmark comparison, and documented experimental workflows**.
- Integrated HD-map geometry, lane context, and trajectory metadata with multi-head attention, improving stability 15% while **troubleshooting experiments, visualizing results, and documenting collaborative research findings** for laboratory review.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and data pipelines using **Python, AWS Glue, S3, SQL, modular interfaces, and structured transformations**, improving response times 30% while reducing accumulated technical debt 20%.
- Automated 20+ unit and integration suites with PyTest, Mockito, Jenkins, and virtual machines, strengthening **data validation, regression coverage, reproducibility, documentation, and troubleshooting** across releases on three platforms.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built React interfaces and eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, metadata schemas, and real-time data updates**.
- Automated Docker and Jenkins releases, reducing deployment time 50% while using **Sentry monitoring, structured logs, regression testing, issue diagnosis, and user-facing workflow analysis** across eight production services.

Skills

Scientific Computing: Python, NumPy, Pandas, SciPy, PyTorch, Time-Series Analysis, Signal Processing, Data Visualization
Data & APIs: FastAPI, Flask, REST APIs, PostgreSQL, SQL, HDF5, Data Modeling, Metadata Management
Research Software: MNE-Python, Jupyter, Simulation, Reproducible Research, Model Evaluation, Documentation, Tutorials, Privacy
Applications: React, TypeScript, PyQt, GUI Development, Client-Server Architecture, Authentication, WebSockets, GitHub
Engineering: Git, PyTest, Mockito, Unit Testing, Integration Testing, Debugging, CI/CD, Docker, Linux, Agile

Projects

Neural Signal Data Workspace | *Python, MNE-Python, NumPy, SciPy, HDF5, FastAPI, PostgreSQL*

- Designed a **research-facing electrophysiology workspace** to import recordings, validate metadata, filter time-series signals, annotate events, query sessions, and expose analysis-ready datasets through documented Python APIs for researchers.

Experiment Control and Monitoring Console | *Python, PyQt, Lab Streaming Layer, SQLite, PyTest*

- Built a **real-time experiment support console** using simulated hardware adapters to configure stimulation protocols, synchronize event markers, monitor signal quality, record operator notes, surface faults, and replay sessions for troubleshooting.

Chat2SQL | *Java, TypeScript, SQL, PostgreSQL, Docker, Maven* | Code

- Built a **schema-aware analytical workspace** supporting 10+ databases through searchable metadata, SQL generation, query execution, schema visualization, and documented cross-platform workflows for flexible access to complex structured datasets.